

COMPETITOR GUIDE - v1.0.0

RANK :: CHALLENGE

SQL Query Challenges

10 CHALLENGES - 2,650 POINTS - SQL ONLY

A progressive, query-writing SQL challenge that takes you from SELECT and WHERE all the way to joins, aggregation, and transactions. Each level unlocks the next. Write the query, earn the points, climb the developer ranking.

Generated 2026 - RANK::CHALLENGE v1.0

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What is Rank Challenge?

Rank Challenge is a query-writing competition for SQL developers. Every level hands you a small task against a sample database. Your job is to write the SQL that answers it - and the platform grades your query automatically.

There are 10 challenges worth a combined 2,650 points. They form a single progression: each level you solve unlocks the next, and your total score places you on the developer ranking.

Every challenge is SQL

The whole track is pure SQL - no application code. You write queries in .sql files inside an in-browser editor. Topics climb from the fundamentals to real reporting:

- Fundamentals - SELECT, FROM, WHERE, choosing columns
- Filtering - AND / OR, parentheses, LIKE and IN
- Joining - INNER JOIN and LEFT JOIN with ON
- Aggregation - COUNT, GROUP BY, HAVING
- Shaping & writing - ORDER BY / LIMIT, UPDATE ... WHERE, transactions

You only need to know SQL

No application, server, or front-end knowledge is required. If you can write a SELECT and a JOIN, you have everything you need.

How grading works

When you press Submit, the server checks your query for the clauses that level requires (for example: "uses a WHERE clause", "joins with INNER JOIN", "groups with GROUP BY"). All checks must pass to clear the level.

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Install Node.js

Rank Challenge runs on Node.js (which includes npm). You need Node 18 or newer; the latest LTS release is recommended.

Step 1 - Check if you already have it

Open a terminal (Terminal on macOS/Linux, or PowerShell / Command Prompt on Windows) and run:

```
node -v  
npm -v
```

If both print a version (and node -v is v18 or higher), you're ready. Otherwise install or update Node below.

Step 2 - Download & install

Official installer (pick LTS): <https://nodejs.org/en/download>

Run the installer, accept the defaults, then reopen your terminal and rerun node -v to confirm.

```
# macOS (Homebrew)  
brew install node  
# Windows (winget)  
winget install OpenJS.NodeJS.LTS  
# Debian / Ubuntu  
sudo apt-get install -y nodejs npm
```

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Run the Challenge

With Node installed, you start the platform from the project folder in three commands.

Step 1 - Open the project

```
cd path/to/sql
```

Step 2 - Install dependencies (first time only)

```
npm install
```

This installs the frontend and backend packages (npm workspaces). It only needs to be done once.

Step 3 - Start it

```
npm run dev
```

This launches the frontend (<http://localhost:5173>) and the backend (<http://localhost:3001>) together.

Step 4 - Open & log in

Open <http://localhost:5173>, click Continue, then enter your 6-digit access code. A successful login starts a 1-hour session.

Login codes are issued by the 2FA service your organiser runs (<http://localhost:3000>). Each code is single-use and expires quickly.

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The 10 Challenges

All ten levels are SQL. Solve them in order - each unlocks the next.

#	TITLE	WHAT YOU'LL FIX	POINTS
01	Select the Columns	SELECT specific columns with FROM	100
02	Filter the Rows	Add a WHERE clause	150
03	Combine Conditions	AND / OR with parentheses	200
04	Join Two Tables	INNER JOIN ... ON	250
05	Count per Group	COUNT() with GROUP BY	250
06	Top Ten Orders	ORDER BY DESC + LIMIT	300
07	Keep Everyone	Switch to a LEFT JOIN	300
08	Update One Row	UPDATE ... SET ... WHERE	350
09	All or Nothing	Wrap writes in BEGIN / COMMIT	350
10	Top Customers Report	JOIN + GROUP BY + HAVING + ORDER BY	400

Each level tells you exactly what to do

Every challenge has a description, a list of objectives, and optional hints. The objectives map directly to the checks the grader runs.

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Scoring & Rules

Points

Each level is worth the points shown above, from 100 (Level 01) up to 400 (the finale) - 2,650 in total. Points are banked the moment all of a level's checks pass.

Hints cost points

Revealing a hint subtracts a fixed penalty from that level's score (around 15 points). Use them only when stuck.

Unlock chain

Levels unlock in sequence: clear Level 01 to open Level 02, and so on to the finale. Your progress is saved within your session.

Session window

A login is valid for 1 hour. When the timer in the top bar runs out you return to the login screen and need a new code.

Submission limit

Submissions are rate-limited to 20 per minute to keep things fair - no penalty, you just wait a moment if you hit it.

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The Interface & Finishing

Top bar

- Server status - confirms the backend is reachable
- Session timer - time left in your 1-hour window
- Score & progress - your points and solved/total count
- Challenge Map - jump back to the level grid
- Finish - end your attempt

Editor & results

The left panel is the query editor (your .sql file). The right panel is a results console - press Submit to grade your query and see which checks passed.

Use Submit to grade your answer, Hint to reveal a hint (costs points), and Reset to restore the original starter query (.sql file).

Finishing the challenge

When you're done, press Finish in the top bar to see a summary of how many levels you solved and your final score, then confirm to end the session.

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Tips & Quick Reference

Strategy

- Read the objectives first - they mirror exactly what the grader checks.
- Remember SQL keywords are case-insensitive - SELECT or select both work.
- Save hints for later - try to fix yourself first; hints cost points.
- Submit early, submit often - failed checks tell you what's still wrong.
- Reset if you get lost - it brings back the original starter.

Quick command reference

```
node -v          # check Node version (need 18+)
npm install      # one-time dependency install
npm run dev      # start frontend + backend
Ctrl + C        # stop both servers
```

Good luck - fix the queries, earn the points, climb the ranking.